
Problem Statement 1

AI-Driven Energy Supply Chain Resilience for Import-Dependent Economies

Theme: Supply Chain Intelligence, Energy Security and Geopolitical Risk

Problem Context

India imports approximately 88% of its crude oil, with 40-45% of that volume transiting the Strait of Hormuz. Recent events - the 2025 US-Iran standoff, renewed sanctions on Iranian exports, and attacks on Red Sea shipping lanes - have repeatedly disrupted supply and pricing, while India's strategic reserves cover only about 9.5 days of national consumption. Existing supply chain planning tools cannot model geopolitical risk in real time or coordinate a rerouting response across refiners, logistics providers and reserves.

Challenge Statement

Design an AI-powered system that continuously monitors geopolitical and logistics risk, models disruption scenarios and their economic impact, and generates executable procurement rerouting recommendations.

Illustrative Directions

- Geopolitical Risk Intelligence Agent - produces a live supply-disruption probability score by corridor and supplier from news, shipping and sanctions data.
- Disruption Scenario Modeller - simulates events such as a Hormuz closure or Red Sea suspension and their cascading impact on refining, prices and GDP.
- Adaptive Procurement Orchestrator - ranks alternative crude sources and logistics routes for procurement teams to act on within hours.
- Strategic Reserve Optimisation Agent - models optimal reserve drawdown schedules against supply gap forecasts.
- Supply Chain Digital Twin - a geospatial simulation of the energy network enabling continuous what-if analysis.

Problem Statement 2

AI for Equitable Education Access

Theme: Education, Language Access and Personalized Learning

Problem Context

Many students lack access to quality tutoring, doubt resolution or personalised feedback - not because content is unavailable, but because no system connects a student's specific confusion to the right explanation, at the right level, in the right language. Teachers in under-resourced schools are similarly stretched, often without a way to identify which students are falling behind until it is too late.

Challenge Statement

Build an AI system that closes part of the education access gap, for students, teachers, or both, using open or public educational content as its knowledge base.

Illustrative Directions

- Grounded Doubt-Solving Agent - explains concepts step by step from open textbooks, with citations to source.
- Adaptive Practice Generator - creates practice questions and explanations tailored to a student's demonstrated gaps.
- Teacher-Facing Insight Agent - flags students needing attention based on quiz and engagement patterns.
- Scholarship or Eligibility Matcher - helps students discover and apply for aid they qualify for.
- Open reinterpretation of the challenge is welcome.

Problem Statement 3

AI for Civic and Legal Empowerment

Theme: Civic Tech, Legal Access and Government Transparency

Problem Context

Citizens often have legitimate rights and entitlements - consumer protections, tenant rights, RTI access, welfare eligibility - that go unused because navigating bureaucratic and legal language is intimidating and time-consuming. Relevant information is scattered across PDFs, notices and portals that are not designed to solve a specific personal problem quickly.

Challenge Statement

Build an AI system that helps a citizen understand and act on their civic or legal rights, translating bureaucratic complexity into a clear, guided path.

Illustrative Directions

- RTI Drafting Agent - converts a plain-language question into a properly formatted application to the right department.
- Rights Navigator - explains, in simple terms, what a person can do about a specific tenant, consumer or workplace dispute.
- Scheme Eligibility Reader - reads government portals and answers eligibility questions in plain language.
- Conversational Form-Filler - interviews the user and auto-populates the official form.
- Open reinterpretation of bureaucracy translation is welcome.

Problem Statement 4

AI Agent Evaluation and Reliability Engine

Theme: Agent Infrastructure, Testing and Failure Prediction

Problem Context

Autonomous AI agents are increasingly deployed for consequential work, yet industry benchmarks report failure on the majority of real-world tasks attempted, with cited rates near 70%. Teams typically ship agents against a handful of manually written test prompts, so real failure modes - tool-call loops, hallucinated confidence, unsafe destructive actions, silent goal drift - surface only after deployment on live data.

Challenge Statement

Build an AI-powered platform that automatically generates realistic and adversarial test scenarios for a given agent, runs it in a sandboxed environment, scores failure modes, and produces a reliability report, functioning as continuous integration for autonomous agents.

Illustrative Directions

- Scenario Generation Engine - reads an agent's tools, prompt and task domain to generate realistic and adversarial test scenarios at scale.
- Sandboxed Execution and Replay Harness - runs the agent against generated scenarios with mocked tools, capturing traces for deterministic replay.
- Failure Mode Classifier - categorises why a run failed, turning raw pass or fail results into an actionable taxonomy.
- Destructive Action Guardrail Tester - probes an agent's willingness to perform irreversible actions under pressure or ambiguous instruction.
- Reliability Scorecard and Regression Tracker - scores and tracks agent reliability across versions and task categories.

Problem Statement 5

AI for Public Good

Theme: Inclusive AI, Social Impact and Empowerment of Underserved Communities

Problem Context

India's rapid adoption of Artificial Intelligence is creating new opportunities across industries, but many of its most underserved communities remain outside this technological transformation. Small and marginal farmers, fishermen, artisans, rural workers, street vendors, persons with disabilities and other vulnerable groups often lack access to timely information, expert assistance, markets and essential public services.

These communities frequently make high-impact decisions with limited resources and incomplete information — whether it is a farmer deciding when and what to cultivate, a fisherman assessing weather and fishing conditions, an artisan finding reliable markets, or a citizen navigating government schemes and services.

The challenge is not simply to make AI more powerful, but to make it more accessible, relevant and useful to the people who need it most.

Challenge Statement

Build an AI-powered solution that addresses a real-world problem faced by an underserved or marginalized community in India, using AI to improve their access to information, decision-making, livelihoods, safety, essential services or economic opportunities.

Solutions should demonstrate a clear understanding of the target community and should be designed with considerations such as local languages, digital literacy, affordability, accessibility and limited connectivity where relevant.

Illustrative Directions

- AI for Farmers — provides crop advisory, disease detection, weather-aware recommendations, market intelligence or assistance in accessing relevant government schemes.
- AI for Fishermen — provides weather and sea-condition intelligence, fishing-zone recommendations, safety alerts, catch prediction or market and supply-chain assistance.
- AI for Artisans and Small Producers — helps discover markets, predict demand, generate product catalogues, improve pricing and connect traditional producers with customers.
- AI for Micro-Entrepreneurs and Street Vendors — assists with demand forecasting, inventory planning, financial decision-making and access to formal markets or government support.
- AI for Accessible Public Services — helps citizens discover, understand and access welfare schemes, government services, documentation and eligibility requirements.
- AI for Persons with Disabilities — develops assistive tools for communication, education, navigation, accessibility or employment.
- AI for Rural Education and Skill Development — provides personalized learning, multilingual educational assistance, career guidance or skill development for students with limited access to quality resources.
- AI for Disaster and Climate Resilience — helps vulnerable communities anticipate, prepare for and respond to floods, droughts, extreme weather and other climate-related risks.
- Open reinterpretation of the challenge is welcome. Participants are encouraged to identify other underserved communities and explore novel applications of AI that can create measurable and sustainable social impact.